

Coordination-Dependent Hydrogen Adsorption on Co-, Fe-, and Ni-Functionalized Monolayer CrCl_3

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Supporting Information Available

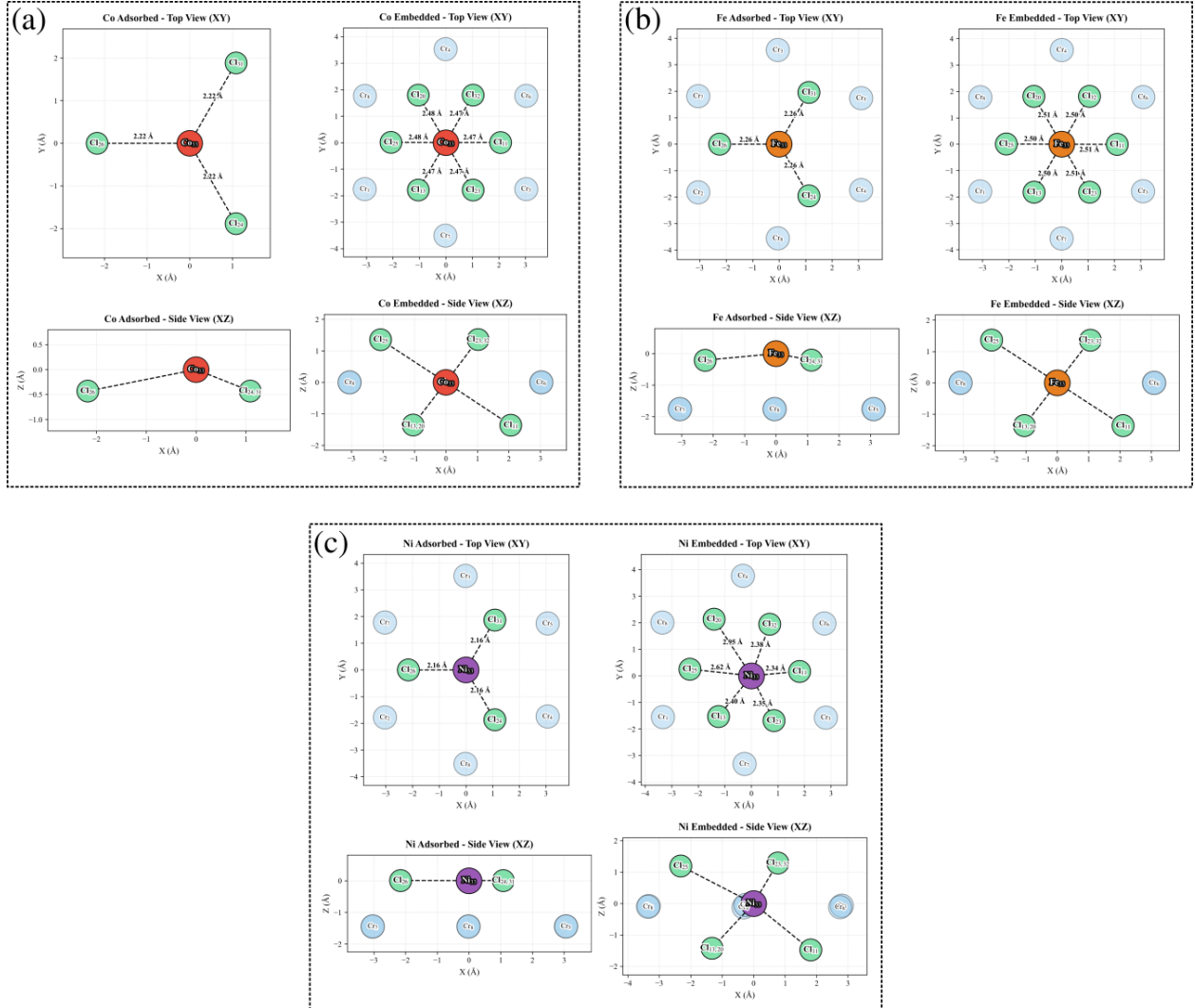


Fig. S1. Schematic top (XY) and side (XZ) views of the optimized local coordination geometries and TM-Cl bond lengths (Å) for (a) Co-, (b) Fe-, and (c) Ni-functionalized monolayer CrCl₃ in both adsorbed (hollow site, 3-fold coordinated) and embedded (6-fold octahedrally coordinated) configurations. Light blue, light green, and red/orange/purple spheres represent Cr, Cl, and transition-metal (Co, Fe, Ni) atoms, respectively.

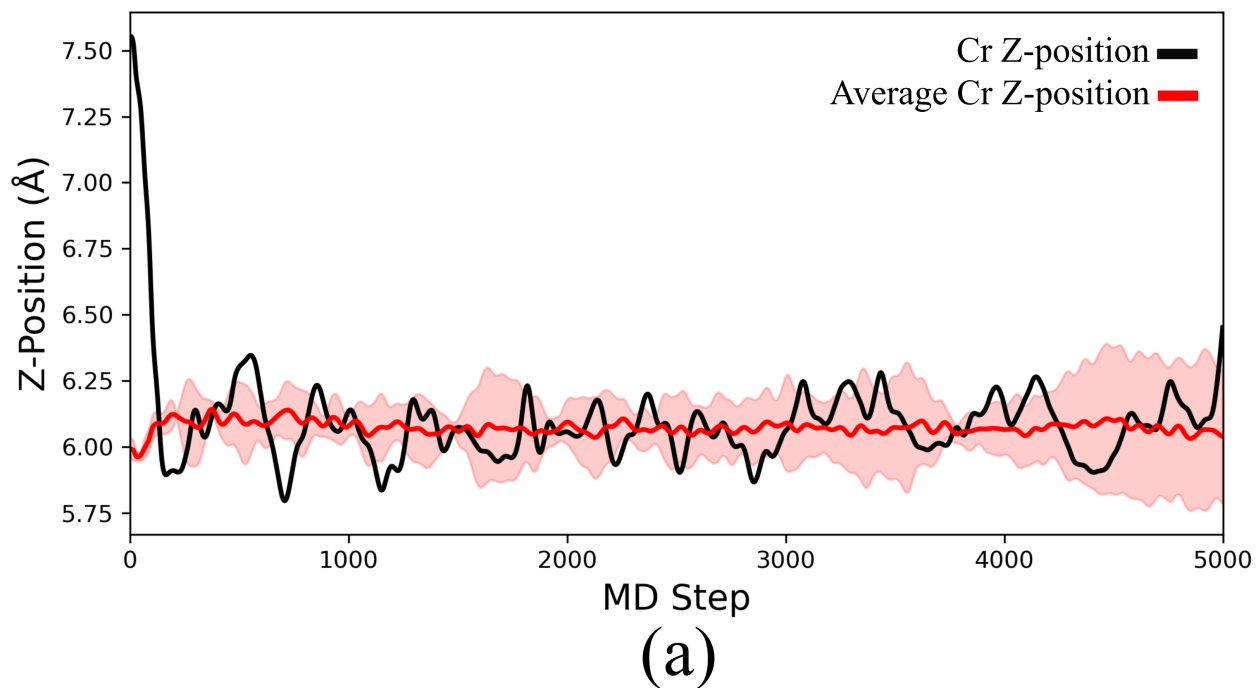


Fig. S2. Time evolution of the absolute z coordinate of the Cr atom initially adsorbed at the hollow site of monolayer CrCl_3 during a 5 ps *ab initio* molecular dynamics simulation at 300 K. The black curve represents the instantaneous z coordinate of the adsorbed Cr atom, whereas the red curve denotes the average z coordinate of the Cr atoms belonging to the host monolayer.

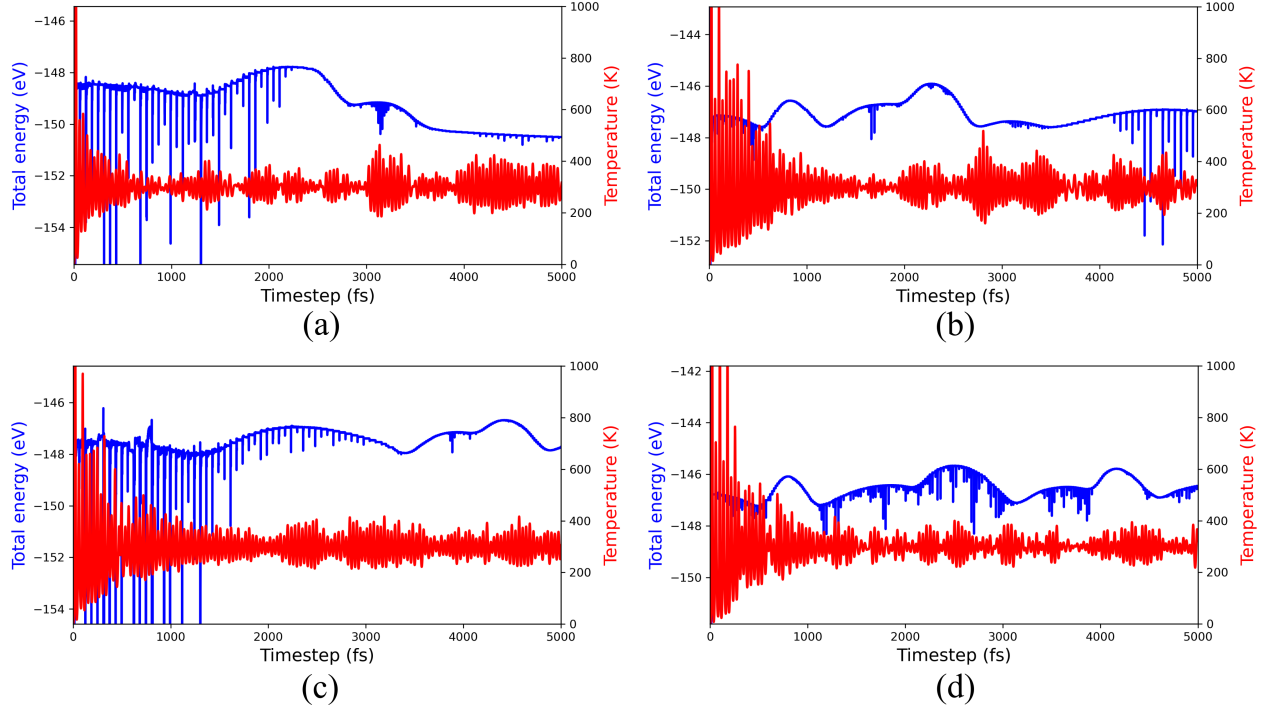


Fig. S3. Time evolution of the total energy (blue curves, left axes) and instantaneous temperature (red curves, right axes) during the 5 ps AIMD simulations at 300 K for monolayer CrCl_3 containing an initially surface-adsorbed (a) Cr, (b) Co, (c) Fe, or (d) Ni atom at the hollow site. Following the initial thermostating transients, the temperatures fluctuate around the target value. The changes in the total-energy profiles accompany the structural rearrangements sampled during the trajectories.

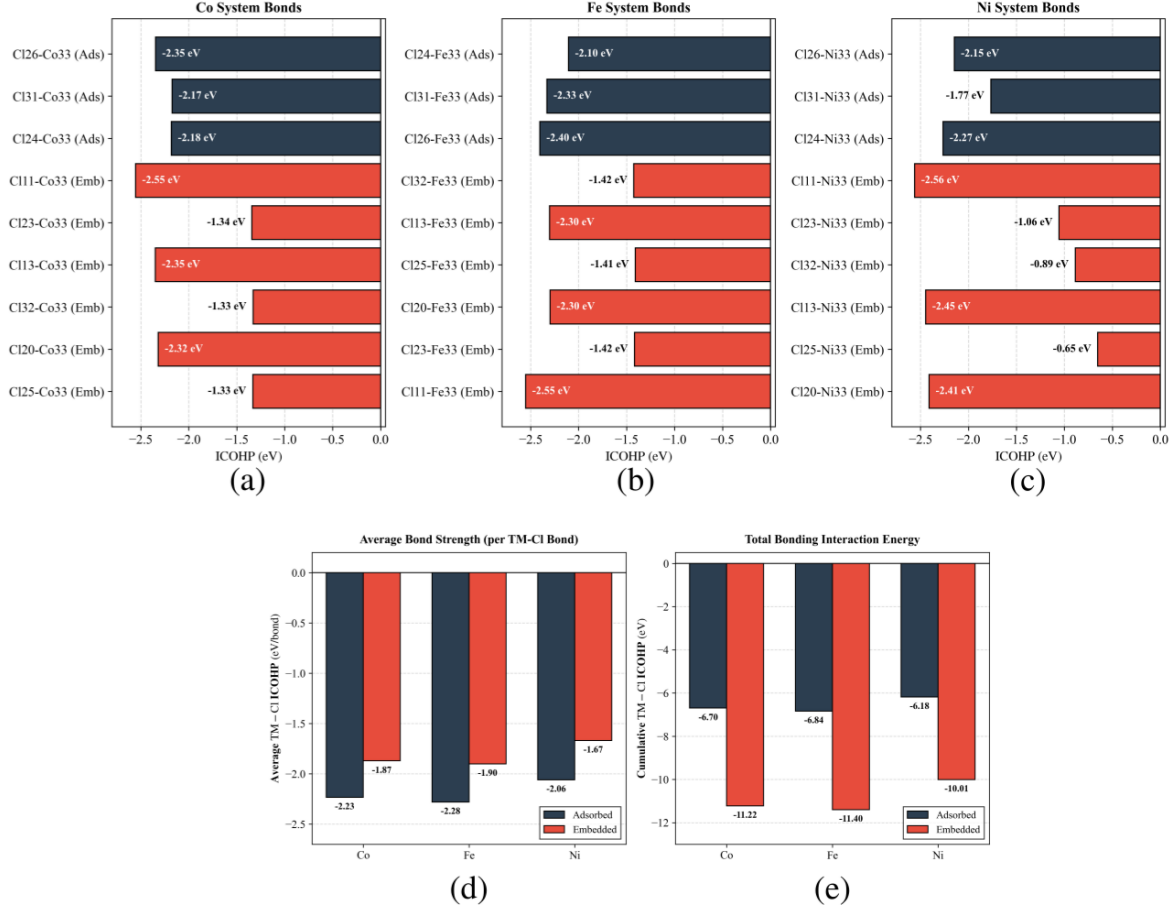


Fig. S4. TM-Cl chemical-bonding analysis and relative stability of surface-adsorbed and embedded transition-metal configurations in monolayer CrCl₃. (a-c) Individual integrated crystal orbital Hamilton population (ICOHP) values for the Co-Cl, Fe-Cl, and Ni-Cl interactions, respectively. Dark bars correspond to the threefold coordinated surface-adsorbed configurations, whereas red bars correspond to the sixfold coordinated embedded configurations. More negative ICOHP values indicate stronger net bonding interactions. (d) Average ICOHP per TM-Cl bond for the adsorbed and embedded configurations. (e) Cumulative TM-Cl ICOHP obtained by summing the individual ICOHP contributions over the corresponding coordination shell.

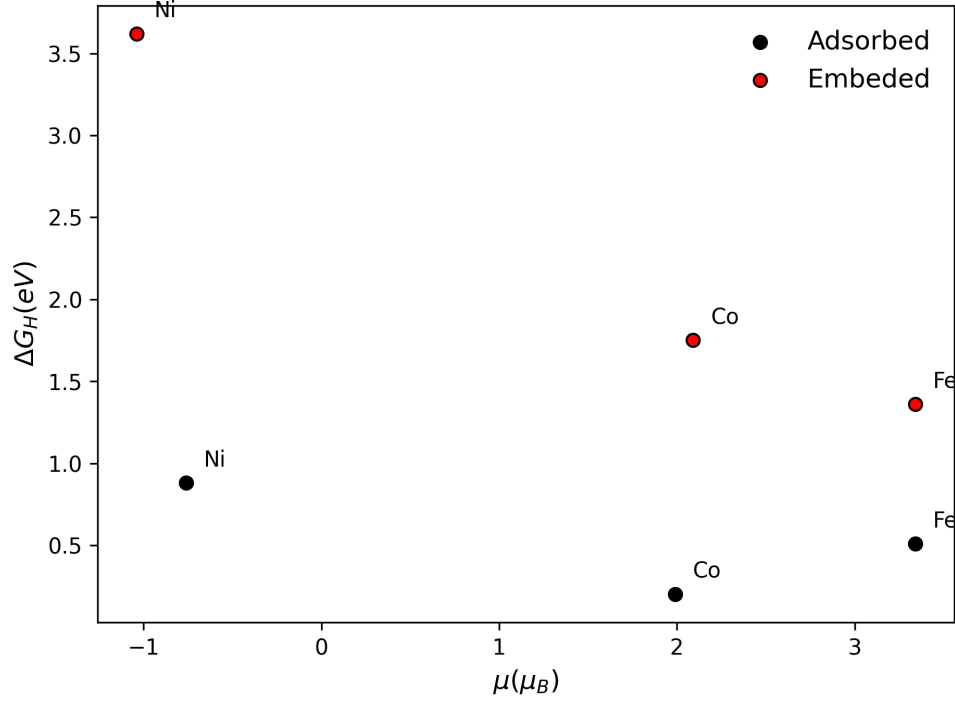


Fig. S5. Relationship between the local magnetic moment of the functionalizing transition-metal atom, μ_{TM} , in the relaxed H-free configurations and the hydrogen adsorption free energy, ΔG_{ads} , subsequently calculated after H adsorption. Black and red symbols represent the surface-adsorbed and embedded Co-, Fe-, and Ni functionalized CrCl_3 monolayers, respectively. Magnetic moments are expressed in units of the Bohr magneton, μ_B .

Supplementary Tables: Comprehensive Structural, Energetic, and Thermodynamic Benchmarks

Table S1: Multi-Scale Hydrogen Adsorption on Pristine Monolayer CrCl_3

Table S1. Comprehensive multi-scale evaluation of hydrogen adsorption on pristine monolayer CrCl_3 across three supercell dimensions (1×1 , 2×2 , and 3×3), three high-symmetry adsorption sites (S1: Top-Cl, S2: Hollow, S3: Top-Cr), and three levels of exchange-correlation dispersion treatment (pure PBE, PBE+D3 zero-damping, and PBE+D3 Becke-Johnson damping). All calculations were performed spin-polarized with $E_{\text{cut}} = 400$ eV and commensurate Monkhorst-Pack k -point grids ($10 \times 10 \times 1$, $5 \times 5 \times 1$, and $3 \times 3 \times 1$, respectively). Adsorption energies are referenced to isolated H_2 gas: $\Delta E_{\text{ads}} = E_{\text{host+H}} - E_{\text{host}} - \frac{1}{2}E(\text{H}_2)$. Free energies are evaluated as $\Delta G_{\text{ads}} = \Delta E_{\text{ads}} + \Delta \text{ZPE} - T\Delta S$, where explicit phononic columns (ΔZPE and $-T\Delta S$) are provided for complete vibrational frequency assignment.

Supercell	Site	Coord.	Method / vdW	E_{host} (eV)	$E_{\text{host+H}}$ (eV)	ΔE_{ads} (eV)	ΔZPE^* (eV)	$-T\Delta S^*$ (eV)	ΔG_{ads} (eV)	μ_{tot} (μ_{B})
1×1 ($\theta = 1.00$)	S1	Top-Cl	PBE (no vdW)	-39.0717	-40.7734	+1.678	+0.040	+0.200	+1.918	7.00
	S2	Hollow	PBE (no vdW)	-39.0717	-39.9647	+2.487	+0.040	+0.200	+2.727	5.04
	S3	Top-Cr	PBE (no vdW)	-39.0717	-40.8155	+1.636	+0.040	+0.200	+1.876	5.00
	S1	Top-Cl	PBE+D3 (zero)	-39.9841	-41.8232	+1.542	+0.040	+0.200	+1.782	7.00
	S2	Hollow	PBE+D3 (zero)	-39.9841	-40.9868	+2.378	+0.040	+0.200	+2.618	5.51
	S3	Top-Cr	PBE+D3 (zero)	-39.9841	-41.7093	+1.656	+0.040	+0.200	+1.896	5.00
	S1	Top-Cl	PBE+D3 (BJ)	-40.5268	-42.5826	+1.325	+0.040	+0.200	+1.565	7.00
	S2	Hollow	PBE+D3 (BJ)	-40.5268	-41.8481	+2.060	+0.040	+0.200	+2.300	7.00
	S3	Top-Cr	PBE+D3 (BJ)	-40.5268	-42.2405	+1.667	+0.040	+0.200	+1.907	5.00
	S1	Top-Cl	PBE (no vdW)	-156.2868	-158.0877	+1.579	+0.040	+0.200	+1.819	25.00
	S2	Hollow	PBE (no vdW)	-156.2868	-157.1957	+2.471	+0.040	+0.200	+2.711	23.00
	S3	Top-Cr	PBE (no vdW)	-156.2868	-158.0422	+1.624	+0.040	+0.200	+1.864	23.00
2×2 ($\theta = 0.25$)	S1	Top-Cl	PBE+D3 (zero)	-159.9366	-161.8036	+1.514	+0.040	+0.200	+1.754	25.00
	S2	Hollow	PBE+D3 (zero)	-159.9366	-160.9571	+2.361	+0.040	+0.200	+2.601	23.00
	S3	Top-Cr	PBE+D3 (zero)	-159.9366	-161.6926	+1.625	+0.040	+0.200	+1.865	23.00
	S1	Top-Cl	PBE+D3 (BJ)	-162.1074	-164.2008	+1.288	+0.040	+0.200	+1.528	25.00
	S2	Hollow	PBE+D3 (BJ)	-162.1074	-163.1641	+2.324	+0.040	+0.200	+2.564	23.01
	S3	Top-Cr	PBE+D3 (BJ)	-162.1074	-163.8525	+1.636	+0.040	+0.200	+1.876	23.00
	S1	Top-Cl	PBE (no vdW)	-351.6468	-353.7608	+1.266	+0.040	+0.200	+1.506	55.00
	S2	Hollow	PBE (no vdW)	-351.6468	-352.5562	+2.470	+0.040	+0.200	+2.710	53.00
	S3	Top-Cr	PBE (no vdW)	-351.6468	-353.4262	+1.600	+0.040	+0.200	+1.840	53.00
3×3 ($\theta = 0.11$)	S1	Top-Cl	PBE+D3 (zero)	-364.7418	-366.8243	+1.298	+0.040	+0.200	+1.538	55.00
	S2	Hollow	PBE+D3 (zero)	-364.7418	-365.7984	+2.324	+0.040	+0.200	+2.564	53.00
	S3	Top-Cr	PBE+D3 (zero)	-364.7418	-366.4930	+1.630	+0.040	+0.200	+1.870	53.00
	S1	Top-Cl	PBE+D3 (BJ)	-364.7418	-366.7699	+1.353	+0.040	+0.200	+1.593	55.00
	S2	Hollow	PBE+D3 (BJ)	-364.7418	-365.7988	+2.324	+0.040	+0.200	+2.564	53.00
	S3	Top-Cr	PBE+D3 (BJ)	-364.7418	-366.4846	+1.638	+0.040	+0.200	+1.878	53.00

*Placeholder zero-point energy ($\Delta \text{ZPE} = +0.040$ eV) and entropic ($-T\Delta S = +0.200$ eV at 298.15 K)

values represent the standard computational hydrogen electrode (CHE) reference terms ($\Delta G = \Delta E_{\text{ads}} + 0.24$ eV). These explicit columns are formatted to receive the full explicit phonon/vibrational frequency mode calculations upon completion by the co-authors.

Table S2: Transition-Metal Functionalization on Monolayer CrCl_3 (2×2 Supercell)

Table S2. Structural energetics, magnetic states, and relative stabilities of transition-metal adatoms (Co, Fe, Ni) functionalized on monolayer CrCl_3 in a 2×2 supercell. Configurations include surface sites: Site 1 (S1, top-Cl, 1-fold coordinated), Site 2 (S2, hollow, 3-fold coordinated), Site 3 (S3, top-Cr, 3-fold coordinated directly above lower Cr), and the embedded configuration (6-fold octahedral coordination in a Cr vacancy). Calculations evaluate standard PBE, PBE+D3(BJ) ($U = 0$), and DFT+U ($U_{\text{eff}} = 3.29$ eV applied exclusively to Cr $3d$). Interaction energy is defined as $E_{\text{int}} = E_{\text{tot}} - E_{\text{host}}$.

Metal (TM)	Site / Coord.	Coord. No.	Level of Theory	E_{host} (eV)	E_{tot} (eV)	E_{int} (eV)	μ_{tot} (μ_B)	Relative Rank	Integrity / Notes
Cobalt (Co)	S1 (Top-Cl)	1	PBE (no vdW)	-156.2868	-161.0147	-4.728	23.00	Metastable ($\Delta E = +0.00$ eV)	Chemisorbed ($d_{\text{Co-Cl}} = 2.14$ Å)
	S2 (Hollow)	3	PBE (no vdW)	-156.2868	-161.0131	-4.726	23.00	Ground State (surface)	Chemisorbed ($d_{\text{Co-Cl}} = 2.24$ Å)
	S3 (Top-Cr)	3	PBE (no vdW)	-156.2868	-160.3660	-4.079	23.00	Metastable ($\Delta E = +0.65$ eV)	Chemisorbed ($d_{\text{Co-Cr}} = 2.59$ Å)
	Embedded	6	PBE (no vdW)	-156.2868	-160.8042	-4.517	27.12	Embedded ground state	Octahedral ($d_{\text{Co-Cl}} = 2.31$ Å)
	S1 (Top-Cl)	1	PBE+D3 (BJ)	-162.1068	-166.9422	-4.836	27.00	Metastable ($\Delta E = +0.22$ eV)	Chemisorbed ($d_{\text{Co-Cl}} = 2.13$ Å)
	S2 (Hollow)	3	PBE+D3 (BJ)	-162.1068	-167.1638	-5.057	23.00	Ground State (surface)	Chemisorbed ($d_{\text{Co-Cl}} = 2.23$ Å)
	S3 (Top-Cr)	3	PBE+D3 (BJ)	-162.1068	-166.4536	-4.347	23.00	Metastable ($\Delta E = +0.71$ eV)	Chemisorbed ($d_{\text{Co-Cr}} = 2.58$ Å)
	S1 (Top-Cl)	1	PBE+D3+U (3.29)	-147.3038	-150.5731 [†]	-3.269	27.00	Metastable ($\Delta E = +1.67$ eV)	Chemisorbed ($d_{\text{Co-Cl}} = 2.16$ Å)
	S2 (Hollow)	3	PBE+D3+U (3.29)	-147.3038	-152.2452 [†]	-4.941	27.00	Ground State (surface)	Chemisorbed ($d_{\text{Co-Cl}} = 2.22$ Å)
	S3 (Top-Cr)	3	PBE+D3+U (3.29)	-147.3038	-151.7732	-4.469	27.00	Metastable ($\Delta E = +0.47$ eV)	Coaxial Cr-Co ferromagnetic ($d_{\text{Co-Cr}} = 2.58$ Å)
	S1 (Top-Cl)	1	PBE (no vdW)	-156.2868	-161.9051	-5.618	28.41	Metastable ($\Delta E = +0.71$ eV)	Chemisorbed ($d_{\text{Fe-Cl}} = 2.16$ Å)
	S2 (Hollow)	3	PBE (no vdW)	-156.2868	-162.6198	-6.333	28.44	Ground State (surface)	Chemisorbed ($d_{\text{Fe-Cl}} = 2.28$ Å)
Iron (Fe)	S3 (Top-Cr)	3	PBE (no vdW)	-156.2868	-161.3136	-5.027	21.65	Metastable ($\Delta E = +1.31$ eV)	Chemisorbed ($d_{\text{Fe-Cr}} = 2.62$ Å)
	Embedded	6	PBE (no vdW)	-156.2868	-162.1081	-5.821	28.96	Embedded ground state	Octahedral ($d_{\text{Fe-Cl}} = 2.36$ Å)
	S1 (Top-Cl)	1	PBE+D3 (BJ)	-162.1068	-168.9135	-6.807	22.00	Ground State (surface)	Chemisorbed ($d_{\text{Fe-Cl}} = 2.14$ Å)
	S2 (Hollow)	3	PBE+D3 (BJ)	-162.1068	-168.1160	-6.009	28.39	Metastable ($\Delta E = +0.80$ eV)	Chemisorbed ($d_{\text{Fe-Cl}} = 2.26$ Å)
	S3 (Top-Cr)	3	PBE+D3 (BJ)	-162.1068	-165.9127	-3.806	24.18	Metastable ($\Delta E = +3.00$ eV)	Chemisorbed ($d_{\text{Fe-Cr}} = 2.61$ Å)
	S1 (Top-Cl)	1	PBE+D3+U (3.29)	-147.3038	[In Queue]	—	—	Staged / In Queue	Auto-monitored
	S2 (Hollow)	3	PBE+D3+U (3.29)	-147.3038	[In Queue]	—	—	Staged / In Queue	Auto-monitored
	S3 (Top-Cr)	3	PBE+D3+U (3.29)	-147.3038	[In Queue]	—	—	Staged / In Queue	Auto-monitored
	S1 (Top-Cl)	1	PBE (no vdW)	-156.2868	-159.7885	-3.502	24.00	Metastable ($\Delta E = +0.29$ eV)	Chemisorbed ($d_{\text{Ni-Cl}} = 2.12$ Å)
	S2 (Hollow)	3	PBE (no vdW)	-156.2868	-160.0785	-3.792	24.00	Ground State (surface)	Chemisorbed ($d_{\text{Ni-Cl}} = 2.21$ Å)
	S3 (Top-Cr)	3	PBE (no vdW)	-156.2868	-159.2846	-2.998	24.00	Metastable ($\Delta E = +0.79$ eV)	Chemisorbed ($d_{\text{Ni-Cr}} = 2.55$ Å)
	Embedded	6	PBE (no vdW)	-156.2868	-159.3190	-3.032	24.00	Embedded ground state	Octahedral ($d_{\text{Ni-Cl}} = 2.28$ Å)
Nickel (Ni)	S1 (Top-Cl)	1	PBE+D3 (BJ)	-162.1068	-165.9142	-3.807	24.00	Metastable ($\Delta E = +0.39$ eV)	Chemisorbed ($d_{\text{Ni-Cl}} = 2.11$ Å)
	S2 (Hollow)	3	PBE+D3 (BJ)	-162.1068	-166.3062	-4.199	24.00	Ground State (surface)	Chemisorbed ($d_{\text{Ni-Cl}} = 2.20$ Å)
	S3 (Top-Cr)	3	PBE+D3 (BJ)	-162.1068	-165.2844	-3.178	24.00	Metastable ($\Delta E = +1.02$ eV)	Chemisorbed ($d_{\text{Ni-Cr}} = 2.54$ Å)
	S1 (Top-Cl)	1	PBE+D3+U (3.29)	-147.3038	[In Queue]	—	—	Staged / In Queue	Auto-monitored
	S2 (Hollow)	3	PBE+D3+U (3.29)	-147.3038	[In Queue]	—	—	Staged / In Queue	Auto-monitored
	S3 (Top-Cr)	3	PBE+D3+U (3.29)	-147.3038	[In Queue]	—	—	Staged / In Queue	Auto-monitored

[†]Intermediate relaxation values undergoing final micro-batch convergence.

Table S3: Hydrogen Evolution Reaction (HER) Adsorption Thermodynamics on TM-Functionalized CrCl_3

Table S3. Hydrogen adsorption thermodynamics, vibrational free energy corrections, and theoretical HER overpotentials (η_{HER}) for single-atom transition-metal (Co, Fe, Ni) catalytic sites on monolayer CrCl_3 . Adsorption energies are evaluated via the computational hydrogen electrode (CHE) framework: $\Delta E_{\text{ads}} = E_{\text{slab+H}} - E_{\text{slab}} - \frac{1}{2}E(\text{H}_2)$. The Gibbs free energy of adsorption is given by $\Delta G_{\text{ads}} = \Delta E_{\text{ads}} + \Delta \text{ZPE} - T\Delta S$. Theoretical overpotentials are defined by the Volmer-step thermodynamic descriptor: $\eta_{\text{HER}} = |\Delta G_{\text{ads}}|/e$. Dedicated columns for ΔZPE and $-T\Delta S$ are explicitly provided to accommodate the co-authors' full vibrational phonon spectrum calculations.

Catalyst System	Site / Geometry	Level of Theory	E_{slab} (eV)	$E_{\text{slab+H}}$ (eV)	ΔE_{ads} (eV)	ΔZPE^* (eV)	$-T\Delta S^*$ (eV)	ΔG_{ads} (eV)	η_{HER} (V)	HER Activity & Rate-Limiting Regime
A. Pristine Monolayer Reference										
Pristine CrCl_3	Top-Cl (S1)	PBE (no vdW)	-156.2868	-158.0877	+1.579	+0.040	+0.200	+1.819	1.819	Inert (Severe Volmer endothermicity)
Pristine CrCl_3	Top-Cl (S1)	PBE+D3 (BJ)	-162.1074	-164.2008	+1.288	+0.040	+0.200	+1.528	1.528	Inert (Severe Volmer endothermicity)
B. Surface-Adsorbed Transition Metals (3-fold Coordinated)										
CrCl_3 -Co (ads)	Atop Co (S2/S3)	PBE (no vdW)	-160.0665	-163.4806	-0.034	+0.040	+0.200	+0.206	0.206	Optimal Catalyst (Near-thermoneutral)
CrCl_3 -Fe (ads)	Atop Fe (S2)	PBE (no vdW)	-161.2266	-164.4956	+0.111	+0.040	+0.200	+0.351	0.351	Moderate Activity (Volmer-limited)
CrCl_3 -Ni (ads)	Atop Ni (S2)	PBE (no vdW)	-159.4708	-162.2285	+0.622	+0.040	+0.200	+0.862	0.862	Suppressed (Weak TM-H binding)
C. Embedded Transition Metals (6-fold Octahedrally Coordinated)										
CrCl_3 -Co (emb)	Atop Co (Oct)	PBE (no vdW)	-160.8042	-162.6794	+1.505	+0.040	+0.200	+1.745	1.745	Poisoned / Suppressed by coordination
CrCl_3 -Fe (emb)	Atop Fe (Oct)	PBE (no vdW)	-162.1081	-164.3669	+1.121	+0.040	+0.200	+1.361	1.361	Poisoned / Suppressed by coordination
CrCl_3 -Ni (emb)	Atop Ni (Oct)	PBE (no vdW)	-159.3190	-161.0585	+1.640	+0.040	+0.200	+1.880	1.880	Severe structural deformation penalty
D. Surface-Adsorbed Centers with Hubbard U ($U_{\text{eff}} = 3.29$ eV) and DFT-D3(BJ) Dispersion										
CrCl_3 -Co (S3)	Atop Co (S3_H)	PBE+D3+U (3.29)	-151.7732	[Computing]	—	+0.040	+0.200	[Pending]	[Pending]	Under active execution (Job 7857)
CrCl_3 -Co (S1)	Atop Co (S1_H)	PBE+D3+U (3.29)	-150.5731	[Chained]	—	+0.040	+0.200	[Pending]	[Pending]	Auto-dispatched upon S1 convergence
CrCl_3 -Co (S2)	Atop Co (S2_H)	PBE+D3+U (3.29)	-152.2452	[Chained]	—	+0.040	+0.200	[Pending]	[Pending]	Auto-dispatched upon S2 convergence
CrCl_3 -Fe	Atop Fe (SX_H)	PBE+D3+U (3.29)	[Queued]	[Queued]	—	+0.040	+0.200	[Pending]	[Pending]	Staged in workflow pipeline
CrCl_3 -Ni	Atop Ni (SX_H)	PBE+D3+U (3.29)	[Queued]	[Queued]	—	+0.040	+0.200	[Pending]	[Pending]	Staged in workflow pipeline

*Reference vibrational benchmarks: zero-point energy ($\Delta \text{ZPE} = +0.040$ eV) and vibrational entropy ($-T\Delta S = +0.200$ eV at 298.15 K) yield the standard +0.240 eV correction. The separate ΔZPE and $-T\Delta S$ columns are specifically reserved for insertion of individual phononic eigenvalues currently being computed by co-authors.